Acetone

1 Nonproprietary Names

BP: Acetone
PhEur: Acetone
USP-NF: Acetone

2 Synonyms

Acetonum; dimethylformaldehyde; dimethyl ketone; β-ketopropane; pyroacetic ether.

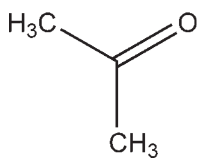
3 Chemical Name and CAS Registry Number

2-Propanone [67-64-1]

4 Empirical Formula and Molecular Weight

C_3H_6O 58.08

5 Structural Formula



6 Functional Category

Solvent.

7 Applications in Pharmaceutical Formulation or Technology

Acetone is used as a solvent or cosolvent in topical preparations, and as an aid in wet granulation.[1,2] It has also been used when formulating tablets with water-sensitive active ingredients, or to solvate poorly water-soluble binders in a wet granulation process. Acetone has also been used in the formulation of microspheres to enhance drug release.[3] Owing to its low boiling point, acetone has been used to extract thermolabile substances from crude drugs.[4]

8 Description

Acetone is a colorless volatile, flammable, transparent liquid, with a sweetish odor and pungent sweetish taste.

9 Pharmacopeial Specifications

See Table I. *See also* Section 18.

Table I: Pharmacopeial specifications for acetone.

Test	PhEur 6.0	USP32–NF27
Identification	+	+
Characters	+	—
Appearance of solution	+	—
Acidity or alkalinity	+	—
Relative density	0.790–0.793	⩽0.789
Related substances	+	—
Matter insoluble in water	+	—
Reducing substances	+	+
Residue on evaporation	⩽50 ppm	⩽0.004%
Water	⩽3 g/L	+
Assay	—	⩾99.0%

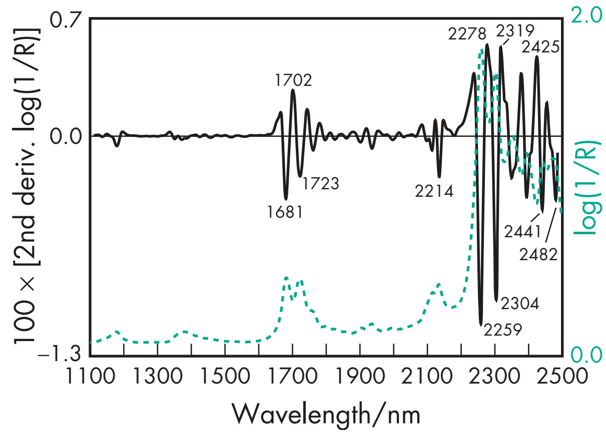


Figure 1: Near-infrared spectrum of acetone measured by transflectance (1 mm path-length).

10 Typical Properties

Boiling point 56.2°C
Flash point –20°C
Melting point 94.3°C
NIR spectra *see* Figure 1.
Refractive index $n_D^{20} = 1.359$
Solubility Soluble in water; freely soluble in ethanol (95%).
Vapor pressure 185 mmHg at 20°C

11 Stability and Storage Conditions

Acetone should be stored in a cool, dry, well-ventilated place out of direct sunlight.

12 Incompatibilities

Acetone reacts violently with oxidizing agents, chlorinated solvents, and alkali mixtures. It reacts vigorously with sulfur dichloride, potassium *t*-butoxide, and hexachloromelamine. Acetone should not be used as a solvent for iodine, as it forms a volatile compound that is extremely irritating to the eyes.[4]

13 Method of Manufacture

Acetone is obtained by fermentation as a by-product of *n*-butyl alcohol manufacture, or by chemical synthesis from isopropyl alcohol; from cumene as a by-product in phenol manufacture; or from propane as a by-product of oxidation-cracking.

14 Safety

Acetone is considered moderately toxic, and is a skin irritant and severe eye irritant. Skin irritation has been reported due to its defatting action, and prolonged inhalation may result in headaches. Inhalation of acetone can produce systemic effects such as conjunctival irritation, respiratory system effects, nausea, and vomiting.[5]

LD_{50} (mouse, oral): 3.0 g/kg[5]
LD_{50} (mouse, IP): 1.297 g/kg
LD_{50} (rabbit, oral): 5.340 g/kg
LD_{50} (rabbit, skin): 0.2 g/kg

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LD_{50} (rat, IV): 5.5 g/kg
LD_{50} (rat, oral): 5.8 g/kg

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Acetone is a skin and eye irritant (*see* Section 14); therefore gloves, eye protection and a respirator are recommended. In the UK, the long-term (8-hour TWA) workplace exposure limit for acetone is 1210 mg/m^3 (500 ppm). The short-term (15-minute) exposure limit is 3620 mg/m^3 (1500 ppm).[6]

16 Regulatory Status

Included in the FDA Inactive Ingredients Database (inhalation solution; oral tablets; topical preparations). Included in the Canadian List of Acceptable Non-medicinal Ingredients. Included in nonparenteral medicines licensed in the UK.

17 Related Substances

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18 Comments

A specification for acetone is included in the *Japanese Pharmaceutical Excipients* (JPE).[7]

The EINECS number for acetone is 200-662-2. The PubChem Compound ID (CID) for acetone is 180.

19 Specific References

1 Ash M, Ash I. *Handbook of Pharmaceutical Additives*, 3rd edn. Endicott, NY: Synapse Information Resources, 2007; 430.
2 Tang ZG *et al*. Surface properties and biocompatibility of solvent-cast poly[ε-caprolactone] films. *Biomaterials* 2004; **25**(19): 4741–4748.
3 Ruan G, Feng SS. Preparation and characterization of poly(lactic acid)–poly(ethylene glycol)–poly(lactic acid) (PLA-PEG-PLA) microspheres for controlled release of paclitaxel. *Biomaterials* 2003; **24**(27): 5037–5044.
4 Todd RG, Wade A, eds. *The Pharmaceutical Codex*, 11th edn. London: Pharmaceutical Press, 1979; 6.
5 Lewis RJ, ed. *Sax's Dangerous Properties of Industrial Materials*, 11th edn. New York: Wiley, 2004; 22–23.
6 Health and Safety Executive. *EH40/2005: Workplace Exposure Limits*. Sudbury: HSE Books, 2005 (updated 2007). http://www.hse.gov.uk/coshh/table1.pdf (accessed 5 February 2009).
7 Japan Pharmaceutical Excipients Council. *Japanese Pharmaceutical Excipients 2004*. Tokyo: Yakuji Nippo, 2004; 35–36.

20 General References

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21 Author

AH Kibbe.

22 Date of Revision

5 February 2009.

Acetyltributyl Citrate

1 Nonproprietary Names

USP-NF: Acetyltributyl Citrate
PhEur: Tributyl Acetylcitrate

2 Synonyms

Acetylbutyl citrate; acetylcitric acid, tributyl ester; ATBC; *Citroflex A-4*; tributyl acetylcitrate; tributylis acetylcitras; tributyl O-acetylcitrate; tributyl citrate acetate.

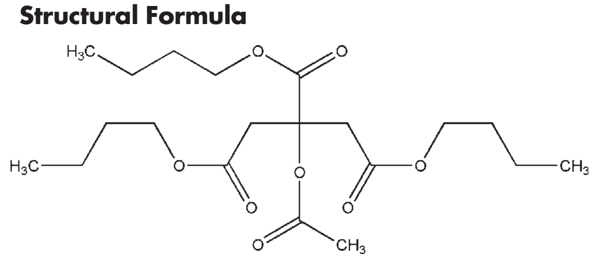
3 Chemical Name and CAS Registry Number

1,2,3-Propanetricarboxylic acid, 2-acetyloxy, tributyl ester [77-90-7]

4 Empirical Formula and Molecular Weight

$C_{20}H_{34}O_8$ 402.5

5 Structural Formula



6 Functional Category

Plasticizer.

7 Applications in Pharmaceutical Formulation or Technology

Acetyltributyl citrate is used to plasticize polymers in formulated pharmaceutical coatings,[1–5] including capsules, tablets, beads, and granules for taste masking, immediate release, sustained-release and enteric formulations.

8 Description

Acetyltributyl citrate is a clear, odorless, practically colorless, oily liquid.

9 Pharmacopeial Specifications

See Table I.

10 Typical Properties

Acid value 0.02
Boiling point 326°C (decomposes)
Flash point 204°C
Pour point −59°C
Solubility Miscible with acetone, ethanol, and vegetable oil; practically insoluble in water.
Viscosity (dynamic) 33 mPa s (33 cP) at 25°C